

Dynamic Fidelity and Model Eligibility in Daily PM₁₀ Forecasting: A Multi-Station Decision Audit

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Abstract

Environmental forecasts can improve on a persistence reference while preserving less of the observed dynamic variation. We examined whether making that distinction explicit changes forecast eligibility in a rolling-origin benchmark of daily PM₁₀ forecasts from 17 Spanish monitoring stations, five model families, and seven horizons. Across 595 structured station–model–horizon cells, Rule A required positive persistence-relative RMSE skill and a significant comparison with persistence; Rule B added operational thresholds for variance retention and P75 exceedance recall. Rule A admitted 277 cells, whereas Rule B admitted 8, so 269 cells changed status (97.1% of Rule-A eligible cells). A threshold sensitivity analysis found that 91.3–98.2% of Rule-A cells changed across the tested neighbourhood. The results show that error and dynamic fidelity can produce different eligibility decisions, while the magnitude is conditional on this pollutant, benchmark, and validation design.

Keywords: PM₁₀ forecasting, model eligibility, dynamic fidelity, forecast evaluation, variance retention, rolling-origin evaluation

1. Introduction

Environmental forecasts are evaluated because decisions depend on future conditions, not only on the average size of past errors. For daily pollution forecasts, the amplitude of episodes, the spread of a predicted trajectory, and sensitivity to elevated values may matter alongside point accuracy. A useful evaluation therefore needs to compare a candidate with a meaningful reference and to describe the properties of the forecast that the error summary does not capture.

Error-based evaluation provides an essential starting point. RMSE measures point-forecast error, while persistence-relative skill asks whether a candidate improves on a simple operational comparator. Predictive-accuracy tests can assess whether the loss difference is statistically distinguishable from that comparator [1, 2, 3]. These tools are necessary for an eligibility decision, but they do not answer every verification question.

A forecast can reduce squared error by attenuating fluctuations. Its error can therefore improve while its variance or exceedance behaviour departs from the observations. Forecast verification treats accuracy, association, calibration, and related properties as complementary dimensions [4, 5]; temporal validation likewise matters when the evaluation is intended to respect deployment order [6, 7]. Variance retention and event recall are used here in that established complementary sense, not proposed as new metrics.

The practical problem begins when several diagnostics are converted into a single eligibility decision. It is not enough to report that fidelity and error are different. The consequential question is whether the set of forecasts judged acceptable changes when dynamic fidelity becomes an explicit requirement alongside error-based skill.

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Existing environmental forecasting studies commonly evaluate predictive error and, less consistently, dynamic-fidelity dimensions [8, 9, 10]. Evidence remains limited on whether explicitly combining these dimensions in an eligibility rule materially changes decisions across sites, model families, and forecast horizons. This bounded gap motivates the decision audit below without making a claim about the entire literature.

We address that question with a post-evaluation analysis of a benchmark of daily PM_{10} forecasts from 17 Spanish monitoring stations, five model families, and seven horizons assessed with five expanding rolling-origin folds. We first identify cells eligible under persistence-relative error skill, then add explicit variance-retention and P75-recall requirements, and finally test whether the decision consequence persists across the prespecified threshold neighbourhood. The contribution is the quantified decision consequence and its bounded evaluation-design implication, not a new metric or a universal eligibility framework.

2. Methods

The study evaluates a benchmark of daily PM_{10} point forecasts and evaluates how eligibility changes when dynamic-fidelity requirements are added to an error-based screen. The analysis is post-evaluation: it does not refit models or optimise a forecasting procedure.

2.1. Data and forecasting task

The evaluation dataset comprises daily PM_{10} series from 17 Spanish monitoring stations identified in MITECO/EMEP metadata. The stations span urban, traffic, industrial, suburban, and rural settings. The recovered station package does not establish one defensible common calendar endpoint for the complete set, so we report verified station coverage and daily resolution rather than infer a period label. The unit of analysis is a station–model–horizon cell; the grid contains 17 stations, five model families, and seven horizons, giving 595 structured cells. These cells share stations and underlying series and are not independent replicates.

The task is direct multi-step daily point forecasting at horizons $h = 1, \dots, 7$. Evaluation follows five expanding rolling-origin folds in chronological order. At each origin, information that became available after that origin was excluded from fitting, feature construction, preprocessing, threshold estimation, and verification inputs. Missing verification targets were excluded rather than fabricated. Exact per-fold row counts and one complete train/test calendar are not recoverable from the aggregate release and are not asserted.

2.2. Forecasting models and persistence reference

The benchmark contains five model families: direct Histogram Gradient Boosting (HGB), direct Ridge regression, fixed-order SARIMA, seasonal naive, and STL+Ridge. HGB and Ridge produce direct forecasts for each horizon. SARIMA is treated as a recursive statistical family, seasonal naive as a seasonal carry-forward reference, and STL+Ridge as a decomposition/regression hybrid. Full verified model-specification details and reconstruction limits are reported in Supplementary Table S2.

Exact HGB, Ridge, and STL settings are not recoverable from the released provenance. Historical SARIMA records conflict; we therefore do not infer an exact order and report only the fixed-order family description supported by the evaluation package. This limits reconstruction detail but does not alter the aggregate diagnostics.

Persistence supplies the operational comparator. It is evaluated on the same matched forecast cases as each candidate, so positive skill means that the candidate reduces RMSE relative to persistence on the relevant cell.

2.3. Evaluation metrics

For matched forecast–observation pairs, RMSE is the square root of mean squared forecast error. Persistence-relative skill is

$$\text{Skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}. \quad (1)$$

Positive skill indicates lower RMSE than persistence; it does not by itself establish dynamic fidelity. The primary dynamic diagnostic is the predicted-to-observed variance ratio,

$$\alpha = \frac{\text{Var}(\hat{y})}{\text{Var}(y)}, \quad (2)$$

computed with population variances over matched cell pairs. P75 recall is the proportion of observed P75 exceedances correctly identified by the forecast, using a threshold estimated from the corresponding training window.

2.4. Eligibility rules and statistical testing

The analysis applies two post-evaluation filters to every station–model–horizon cell. They quantify cell eligibility; they do not select one final winner for each station and horizon.

$$\textbf{Rule A:} \quad \text{Skill} > 0 \quad \text{and} \quad \text{DM}_{BH} \text{ significant}. \quad (3)$$

The two-sided Diebold–Mariano comparison uses the Harvey–Leybourne–Newbold small-sample correction and horizon-dependent autocovariance truncation. Benjamini–Hochberg adjustment is applied to valid model–horizon p-values within each dataset before significance is assigned at $p < 0.05$.

$$\textbf{Rule B:} \quad \text{Rule A} \quad \text{and} \quad \alpha \geq 0.50 \quad \text{and} \quad \text{Recall}_{P75} \geq 0.20. \quad (4)$$

The alpha and P75-recall thresholds are operational criteria for this analysis, not estimated optima, universal standards, or regulatory recommendations. Formal error–fidelity discordance is narrower than the geometric region in Figure 2: a cell must pass Rule A, have P75 recall of at least 0.20, and have $\alpha < 0.50$.

2.5. Threshold sensitivity and reproducibility

The threshold analysis is a deterministic transformation of the analysis rows. It changes neither forecasts nor the primary thresholds. We evaluate the prespecified neighbourhood $\alpha \in \{0.40, 0.50, 0.60\}$ and $\text{Recall}_{P75} \in \{0.10, 0.20, 0.30\}$ to assess whether the eligibility consequence persists; this is not a threshold-optimisation search.

The canonical diagnostic table is the sole numerical source for the manuscript’s results and primary figures. Deterministic generators, provenance records, and claim-level traceability are retained in the repository package, while detailed model specification and sensitivity tables are provided in Supplementary Material.

3. Results

The results move from conventional error eligibility to the dynamic properties of those eligible cells, then to the decision consequence of combining both requirements. This order matters: the reduction from 277 to 8 cells is meaningful only after the two dimensions have been shown to disagree.

3.1. Error skill and dynamic fidelity diverge across model families

Rule A identifies a substantial set of cells that improve on persistence: 111 HGB, 117 Ridge, 44 SARIMA, 5 seasonal-naive, and no STL+Ridge cells. These are eligibility counts within the benchmark, not population rankings and not evidence that one model is the final winner for every station and horizon. Table 1 gives the corresponding family medians and significance counts.

The families that contribute most of the error-eligible cells also occupy the low-alpha part of the diagnostic space. Alpha is below 0.50 in 118 of 119 HGB cells, 118 of 119 Ridge cells, and 110 of 119 SARIMA cells. By contrast, seasonal naive has no low-alpha cells, while STL+Ridge also has none but contributes no Rule-A cells. Figure 1 shows how these contrasts persist or evolve from horizon 1 through 7: skill and retained variance are not interchangeable summaries of forecast quality. The patterns are descriptive and do not establish an architectural or causal mechanism.

Table 1: Per-model diagnostic summary across the 595 structured station–model–horizon cells. Skill and α are medians; all other entries are counts. Cells share stations and series and are not independent.

Model	n	Median Skill	Median α	Collapse	DM sig.	Rule A	Rule B	Discordant
HGB direct	119	0.205	0.151	118/119	111/119	111/119	1/119	54/119
Ridge direct	119	0.219	0.087	118/119	117/119	117/119	1/119	43/119
SARIMA	119	0.208	0.095	110/119	44/119	44/119	1/119	4/119
Seasonal naive	119	-0.026	1.000	0/119	39/119	5/119	5/119	0/119
STL+Ridge	119	-1.107	1.399	0/119	119/119	0/119	0/119	0/119

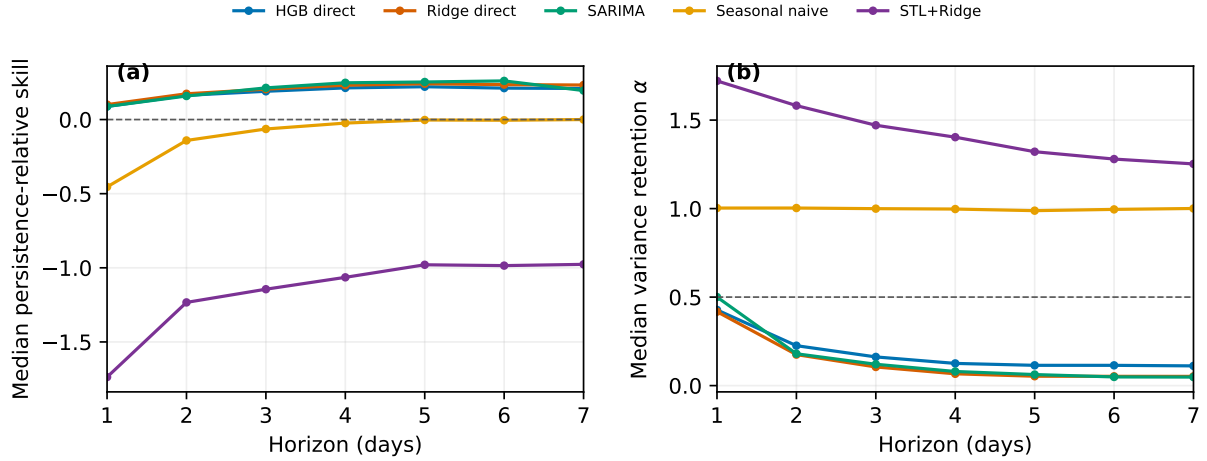


Figure 1: Median persistence-relative RMSE skill and median variance retention by forecast horizon and model family. Each point summarises the 17 station cells at that horizon; the reference lines mark zero skill and $\alpha = 0.50$.

3.2. Error–fidelity discordance appears within the error-eligible space

The cell-level diagnostic makes the family contrast more precise. Figure 2 places persistence-relative skill against variance retention using common axes for the five families. The shaded area marks only positive skill with $\alpha < 0.50$; the open contours mark the stricter formal discordance definition, which also requires Rule A and P75 recall of at least 0.20.

There are 101 formal discordant cells: 54 HGB, 43 Ridge, and 4 SARIMA. Thus, positive error eligibility and adequate P75 recall can coexist with low retained variance in identifiable station–model–horizon cells. The pooled Spearman association between alpha and skill is $\rho = -0.863$, but this is descriptive: it partly reflects the separation among model-family profiles visible in the facets and should not be read as a homogeneous within-family law.

3.3. Fidelity requirements materially change model eligibility

The joint evidence changes the decision set when fidelity is made an explicit requirement. Rule B retains one HGB, one Ridge, one SARIMA, five seasonal-naive, and no STL+Ridge cells, compared with 277 cells under Rule A. The total therefore falls from 277 to 8, with 269 cells changing status; 269/277 is 97.1% of the Rule-A eligible cells. Table 2 puts that transition at the centre of the comparison.

The natural objection is that this consequence may depend on one operational threshold pair. The threshold sensitivity analysis addresses that objection without refitting forecasts. Across α thresholds of 0.40, 0.50, and 0.60 and P75-recall thresholds of 0.10, 0.20, and 0.30, 91.3–98.2% of Rule-A cells change status. The primary point remains 8 retained cells and 97.1% changed. Figure 3 shows the tested neighbourhood; exact retained counts are provided in Supplementary Table S3. The result supports persistence of the qualitative decision consequence, not threshold optimality or universal robustness.

Error-fidelity structure by model family

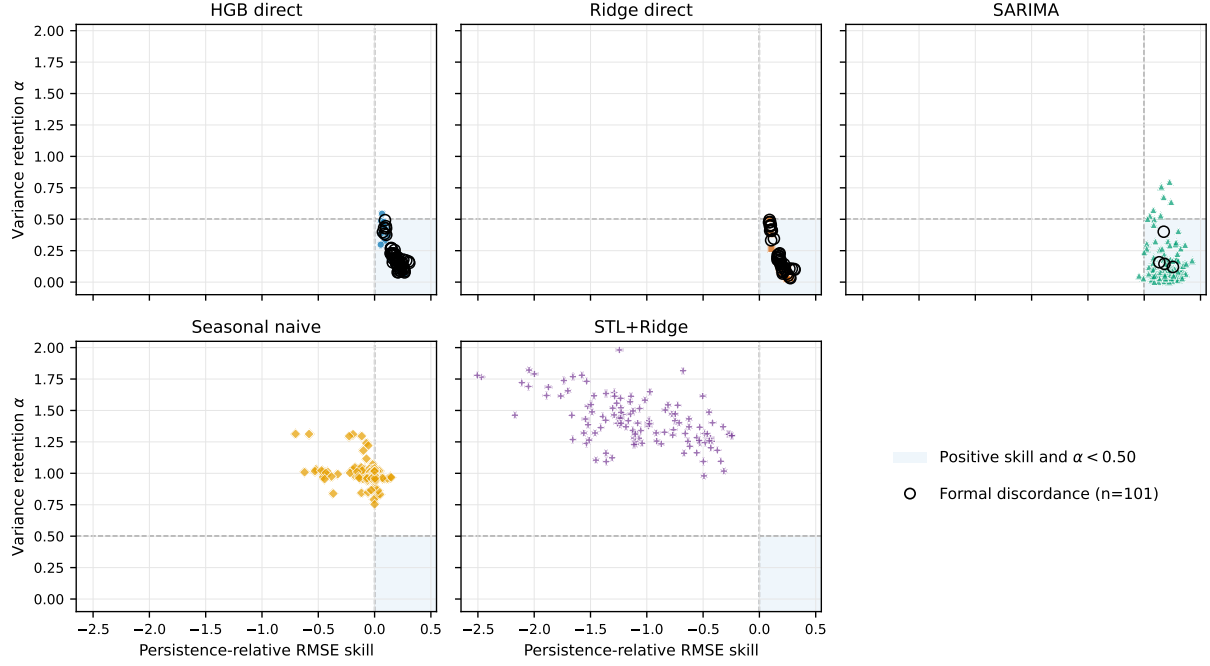


Figure 2: Persistence-relative RMSE skill versus variance retention α across 595 station-model-horizon cells, shown in common-axis facets by model family. The shaded region denotes only positive skill with $\alpha < 0.50$; it is not the formal discordance definition. Open black contours identify the 101 cells that additionally satisfy Rule A and $P_{75} \text{ recall} \geq 0.20$. Dashed lines mark Skill = 0 and $\alpha = 0.50$.

Table 2: Effect of adding dynamic-fidelity requirements (Rule B) to error-based eligibility (Rule A) across the 595-cell empirical benchmark. Rule A: skill > 0 and BH-adjusted DM significant; Rule B: Rule A and $\alpha \geq 0.5$ and $\text{recall}_{P_{75}} \geq 0.20$. Decision changes indicate the number of cells eligible under Rule A but not Rule B. Cells are not independent observations.

Model	Rule A	Rule B	Changes
HGB direct	111	1	110
Ridge direct	117	1	116
SARIMA	44	1	43
Seasonal naive	5	5	0
STL+Ridge	0	0	0
Total	277	8	269 (97.1%)

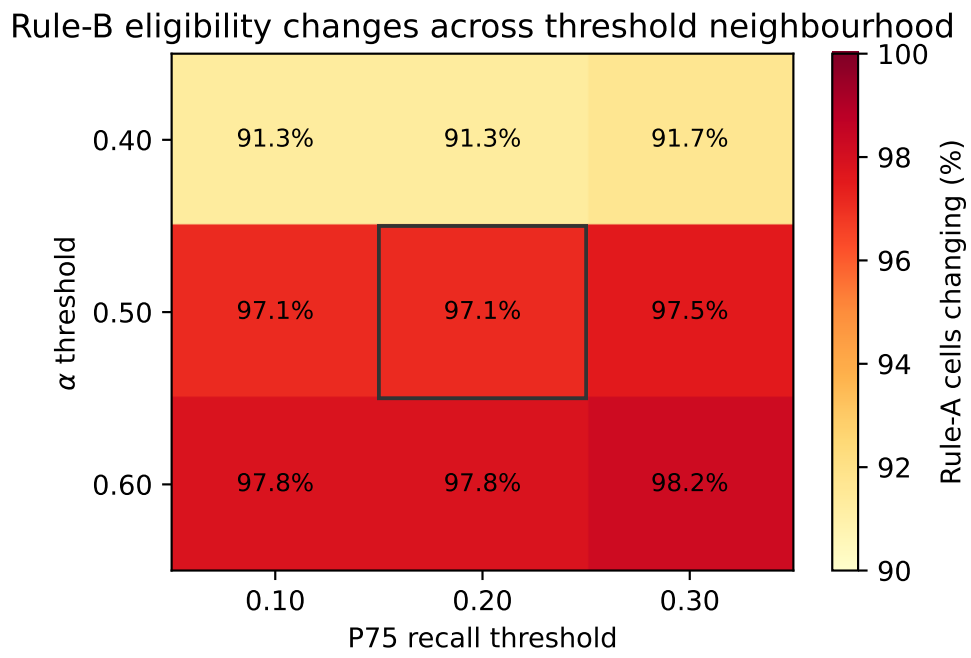


Figure 3: Deterministic sensitivity of the percentage of Rule-A eligible cells that change status when the Rule-B alpha and P75-recall thresholds vary over the tested neighbourhood. The primary audit point is $\alpha = 0.50$ and P75 recall = 0.20; the grid is a sensitivity display, not an optimisation search.

4. Discussion

4.1. What changes when fidelity enters environmental model evaluation

The eligibility analysis revealed a large separation between forecasts that improved on persistence and forecasts that also reproduced the selected dynamic properties of the observed PM_{10} series. Rule A admitted 277 cells, but adding the stated alpha and P75-recall requirements left 8; the 101 formal discordant cells show why the change is not merely a difference between family-level averages. Error skill and dynamic fidelity can therefore lead an evaluation toward different eligibility decisions.

The thresholds are operational criteria for this analysis, and the 595 cells share stations and underlying series. The 97.1% figure therefore describes the proportion of Rule-A cells that changed in this benchmark; it is not a population rate.

RMSE remains an appropriate and necessary measure of point-forecast error. The analysis does not reject it. Rather, it separates the question of error magnitude from the question of whether a forecast retains the variance and exceedance behaviour that may matter for a particular use. A forecast can reduce many squared errors while attenuating its trajectory, while another can retain spread but mistime or mis-size events. Treating those properties as complementary requirements makes the reason for a cell's eligibility decision more explicit.

The model-family profiles reinforce this interpretation without identifying a causal architectural mechanism. HGB, Ridge, and SARIMA contribute many Rule-A cells but also show frequent low-alpha cells. Seasonal naive retains variance in this diagnostic while contributing few error-eligible cells; STL+Ridge avoids low alpha at the point-estimate level but has poor persistence-relative skill and contributes no Rule-A cells. Different evaluation dimensions consequently reward different forecast properties, and a family that looks favourable on one dimension need not be favourable on the joint eligibility question.

4.2. Interpretation, transferability and limits

The result sits within an established evaluation conversation rather than introducing a new metric. Baseline-relative accuracy comparison, predictive-accuracy testing, temporal validation, and multidimensional forecast verification are all established practices [1, 2, 3, 6, 7, 4, 5]. Air-quality forecasting reviews likewise show why error summaries remain a central part of environmental model assessment [8, 9, 10]. The narrower contribution here is to quantify the decision consequence of placing explicit dynamic-fidelity conditions after conventional error-based eligibility across a structured multi-station, multi-family, multi-horizon benchmark.

Because the 97.1% result depends on operational cut-offs, a moderate change in those cut-offs could in principle remove the apparent decision consequence. It does not disappear across the evaluated 3 x 3 neighbourhood: 91.3–98.2% of Rule-A cells change status. This supports a robust qualitative conclusion within the tested neighbourhood, not the optimality of alpha 0.50 or P75 recall 0.20 and not universal robustness of Rule B.

The transferable insight is methodological rather than empirical: environmental model-evaluation conclusions may change when baseline-relative error performance and preservation of decision-relevant dynamics are treated as distinct requirements. That principle may be useful when forecasts inform episode detection, threshold decisions, or amplitude-sensitive planning, but this benchmark does not measure the value of any specific intervention. The 97.1% change, the 277-to-8 transition, and the family-specific collapse rates remain properties of this PM_{10} case.

The analysis concerns daily PM_{10} from 17 Spanish monitoring stations, five model families, seven horizons, deterministic point forecasts, and five expanding rolling-origin folds. The station–model–horizon cells are structured rather than independent, Rule-B thresholds are study-specific, and no causal mechanism for low alpha is established. Historical SARIMA configuration records conflict, so the exact order is not asserted and reconstruction is limited by the recovered aggregate provenance. Transfer to other pollutants, countries, temporal resolutions, horizons, or model families requires a separate evaluation.

5. Conclusions

In this multi-station daily PM₁₀ benchmark, conventional error-based eligibility and dynamic fidelity identified different sets of forecast cells. Rule A admitted 277 cells, whereas Rule B admitted 8; 269 cells changed status, and 101 cells met the formal discordance definition. The threshold sensitivity analysis showed that the qualitative eligibility consequence persisted across the evaluated neighbourhood around the primary cut-offs.

The methodological implication is bounded but transferable: environmental model evaluation should make clear whether error improvement and preservation of decision-relevant dynamics are separate requirements. The observed percentages and eligibility counts belong to this pollutant, station network, model set, horizon range, and validation design. They do not establish a universal rate or mechanism for other forecasting systems.

CRediT author statement

Federico García Crespi: Conceptualisation, Data curation, Formal analysis, Investigation, Methodology, Software, Visualisation, Writing – original draft, Writing – review & editing. **Julio Alberto Ramos Martínez:** Conceptualisation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Software Availability

The analysis repository is maintained at <https://github.com/fedeg-umh-es/varret-pm10-paper>. The canonical aggregate diagnostic table, provenance manifest, reproducible table and figure generators, and documented reconstruction limitations are included in the repository package. The empirical evidence identifies 17 Spanish MITECO/EMEP monitoring stations and contains 595 station–model–horizon rows. The repository must be synchronised with the final submission commit before an EMS submission; no external DOI is asserted here.

FINAL_REPOSITORY_SYNC_REQUIRED = YES.

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The authors used a generative AI assistant for language editing, code/document assistance, and L^AT_EX formatting. No data, forecasts, empirical results, scientific conclusions, or submitted artwork were generated or altered by generative AI. Scientific figures are deterministic plots generated from the analysed data with conventional plotting software; no generative-image tool was used. All data processing, analysis, results, and scientific conclusions are the authors' own and were verified by the authors, who take full responsibility for the content of this paper. This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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Supplementary Material

S1. Monitoring stations

Table 3: Monitoring stations represented in the recovered station metadata. The table is descriptive and does not introduce additional empirical rows beyond the canonical 595-cell benchmark.

Station code	Name	Area type	Altitude (m a.s.l.)	Coverage (%)
ES0446A	Elx-Agroalimentari	Suburban/Industrial	44	80.4
ES1619A	Valencia-Vivers	Urban/Residential	11	91.7
ES0012R	Zarra-EMEP	Rural Remote EMEP	855	96.0
ES1870A	Barcelona El Port Vell	Suburban/Industrial	1	98.5
ES1418A	Alagon	Suburban/Traffic	235	97.5
ES1421A	Teruel	Urban/Background	915	96.7
ES0001R	San Pablo de los Montes	Rural	923	96.1
		Remote/Background		
ES1438A	Barcelona L'Eixample	Urban/Traffic	26	95.3
ES0567A	Barcelona Zona Universitaria	Urban/Background	24	95.2
ES1417A	Huesca	Urban/Traffic	488	95.0
ES0691A	Barcelona El Poblenou	Urban/Background	3	94.9
ES0559A	Barcelona Pl. Universitat	Urban/Traffic	24	94.9
ES1856A	Barcelona Vall d'Hebron	Urban/Background	136	94.5
ES1966A	Alcaniz Capuchinos	Urban/Industrial	340	94.3
ES2011A	Sant Vicenc dels Horts	Suburban/Industrial	65	94.1
	Alaba			
ES2091A	Alcanar Montecarlo	Rural	6	93.7
		Remote/Industrial		
ES2017A	Alcanar Llar de Jubilats	Rural	7	93.6
		Near-city/Industrial		

S2. Model evaluation specification

S3. Rule-B threshold sensitivity

Supplementary Table S2. Model evaluation specification. Reconstructed from the analysis provenance. “Not recoverable” denotes a detail not established by the provenance; it is not inferred from model names.

Model	Inputs	Forecast strategy	Fold fitting	Verified configura- tion	Known limitation
HGB direct	Direct lag-based predictors; complete feature list not recoverable	One direct forecast per horizon	Fit within rolling-origin training data	Family label and 7-horizon evaluation verified	Exact hyper-parameters not recoverable from frozen provenance
Ridge direct	Direct lag-based predictors; complete feature list not recoverable	One direct forecast per horizon	Fit within rolling-origin training data	Family label and 7-horizon evaluation verified	Exact regularisation and feature list not recoverable
SARIMA	Daily PM ₁₀ series	Recursive statistical forecast	Fit within rolling-origin training data	Fixed-order SARIMA family verified	Historical configuration records omitted
Seasonal naive	Historical seasonal observations	Seasonal carry-forward baseline	No parameter fitting beyond training history	Family label and 7-horizon evaluation verified	Seasonal period details not recoverable from canonical table
STL+Ridge	Decomposition and regression inputs	Hybrid decomposition/regression forecast	Fit within rolling-origin training data	Family label and 7-horizon evaluation verified	Exact STL and Ridge settings not recoverable

Supplementary Table S3. Each cell reports retained Rule-B eligible cells and, in parentheses, the percentage of Rule-A cells that change status. The 595 canonical diagnostic rows are reused; no forecasts are recomputed.

α threshold \ recall threshold	0.10	0.20	0.30
0.40	24 (91.3%)	24 (91.3%)	23 (91.7%)
0.50	8 (97.1%)	8 (97.1%)	7 (97.5%)
0.60	6 (97.8%)	6 (97.8%)	5 (98.2%)